P.S.R. ENGINEERING COLLEGE, SIVAKASI – 626 140

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
QUESTION BANK

Programme:	B.E.	Branch	Electronics and Communication Engineering
Academic Year:	2025-2026	Year/Semester	II/IV
Course Code:	231EC45	Course Name	JAVA Programming
Course Tutor:	Mr.D. Arun Shunmugam, AP/ECE Ms.A. Madhumathi AP/ECE		

UNIT- III I/O PACKAGE

PART – A			
1.	Which of this class is used to read characters in a file? a) FileReader b) File Writer c) FileInputStream d) InputStreamReader Answer: a	K1	CO3
2.	Which of this stream contains the classes which can work on character stream? a) InputStream b) OutputStream c) Character Stream d) All of the mentioned Answer:c	K1	CO3
3.	What type of data can be read by an InputStream? a) Characters b) Raw bytes c) Strings d) Objects Answer:b	K1	CO3
4.	_____is a process of extracting/removing the state of an object from a stream. a) Serialization b) Externalization c) File Filtering d) Deserialization Answer:d	K1	CO3
5.	Which of this interface extends DataInput interface? a) Serializable b) Externalization c) ObjectOutput d) ObjectInput Answer:d	K1	CO3
6.	State whether the following statements about the stream in Java. i) The two basic streams used are the input and output streams. ii) Filters are used to read data from one stream and write it to another stream. A) True, True B) True, False C) False, True D) False, False Answer:a	K1	CO3
7.	_____ streams provide functionally for threads to communicate and exchange data between them. A) Object B) Piped C) Pushback D) Filtered Answer:b	K1	CO3

8.	Multithreaded programming is a process in which _____ a) two different processes run simultaneously b) two or more parts of same process run simultaneously c) many different process are able to access same information d) single process can access information from many source Answer:b	K1	CO3
9.	Thread priority in Java is _____ a) Integer b) Float c) double d) long Answer:a	K1	CO3

10.	What is the name of the thread in the following Java Program?
	
	`class multithreaded_programing`
	`{`
	`public static void main(String args[])`
	`{`
	`Thread t = Thread.currentThread();`
	`System.out.println(t);`
	`}`
	`}`
	
	a) main b) Thread c) System d) None of the mentioned
	
	Answer: b
		K1	CO3

11.	Which method is used to start a thread in Java? a) `run()` b) `start()` c) `execute()` d) `init()` Answer: b	K1	CO3
12.	Which of the following is a disadvantage of multi-threading? a) Improved performance b) Reduced CPU utilization c) Complexity in debugging and testing d) Simpler code structure Answer:c	K1	CO3
13.	What is the primary advantage of multi-threading in Java? a) Better resource management b) Simultaneous execution of tasks c) Enhanced debugging capabilities Answer: b d) Decreased memory usage	K1	CO3
14.	Which of the following best describes the life cycle of a thread in Java**?** a) New -> Ready -> Running -> Waiting -> Terminated b) New -> Blocked -> Runnable -> Terminated c) Ready -> Running -> Waiting -> Blocked -> Terminated d) New -> Waiting -> Terminated -> Blocked Answer: a	K1	CO3
15.	What is the potential issue with thread synchronization in multi-threading?	K1	CO3

	a) Deadlock b) Parallelism c) Thread starvation d) Faster execution Answer: a		
16.	In multithreading, which method allows a thread to give up CPU and let other threads run? a) sleep() b) yield() c) wait() d) join() Answer: b	K1	CO3
17.	Which class is used to read primitive data types from a stream? a) ObjectInputStream b) DataInputStream c) FileReader d) BufferedReader Answer: b	K1	CO3
18.	Which of the following is true about **FileWriter** in Java? a) Reads character streams b) Writes character streams c) Writes byte streams d) Reads byte streams Answer: b	K1	CO3
19.	Which exception is thrown when a thread is waiting, sleeping, or otherwise paused for a long time and another thread interrupts it? a) InterruptedException b) IOException c) RuntimeException d) FileNotFoundException Answer: a	K1	CO3
20.	Which method is used to pause the execution of a thread for a specified time? a) wait() b) sleep() c) yield() d) stop() Answer: b	K1	CO3

PART-B (2 marks)			
SL.NO	**QUESTION**	**BTL**	**CO**
1.	What is a stream? Explain its types.	K1	CO3
2.	List classes that are used by Byte streams for input and output operation.	K1	CO3
3.	List classes that are used by character streams for input and output operation	K1	CO3
4.	What is race condition?	K1	CO3
5.	Define Object Serialization.	K1	CO3
6.	Define Serialization and Deserialization in java with example.	K2	CO3
7.	What is multithreading? State its advantages.	K1	CO3
8.	What is thread?	K1	CO3
9.	Compare thread,process and task.	K2	CO3
10.	Outline thread lifecycle.	K1	CO3
11.	What is the difference between InputStream and OutputStream in Java?	K1	CO3
12.	Explain the concept of object serialization in Java.	K2	CO3
13.	What is the use of ObjectOutputStream and ObjectInputStream?	K1	CO3
14.	Define deserialization. How is it different from serialization?	K1	CO3
15.	What is the purpose of a FileReader class in Java?	K2	CO3
16.	How can you copy a file using Java I/O classes?	K1	CO3
17.	Explain the use of Filter streams in Java I/O.	K2	CO3
18.	What are Pipe Streams in Java?	K1	CO3
19.	What is the function of the BufferedReader class in Java I/O?	K2	CO3
20.	How can you read data from a file using a Scanner class in Java?	K1	CO3

SL.NO	QUESTION	BTL	CO
1.	Explain the concepts of Input Streams and Output Streams in Java with suitable examples. Also, discuss object serialization and deserialization, providing a sample program demonstrating object serialization and deserialization to a file.	K3	CO3
2.	Describe the life cycle of a thread in Java. Discuss the advantages and common issues associated with multi-threading. Write a simple Java program to create and run a thread, and explain how thread synchronization can be implemented to avoid thread interference.	K3	CO3
3.	Describe the **life cycle of a thread** in Java. Explain the different states a thread can be in, and the transitions between these states with an example or diagram.	K2	CO3
4.	Explain the concept of **Input Streams** and **Output Streams** in Java. Discuss the various classes used for input and output operations in Java and provide a sample program demonstrating how to read and write data from/to files.	K2	CO3
5.	Explain the concept of **Filter Streams** and **Pipe Streams** in Java. Provide an example of how they are used for filtering or piping data between threads.	K3	CO3
6.	Discuss the **advantages** and **issues** associated with using multi-threading in Java. Explain how multi-threading can improve performance and the potential problems that may arise, such as thread interference.	K2	CO3
7.	Develop a Swing program with a JFrame containing a JButton and a JTextField. When the button is clicked, the text in the text field should be displayed in a JLabel.	K3	CO3
8.	Develop a Java program demonstrating the use of FlowLayout and GridLayout. Place multiple buttons in a frame and show how the layout manager affects their placement.	K3	CO3
9.	Construct a Java applet that reads a numeric parameter from HTML and displays its square on the applet window. Build a simple AWT applet that draws a rectangle and a circle on the screen.	K3	CO3